

Mechanical Engineering Experiments (II)-Worksheet for Project Meeting

機械工程實驗(下)-專題討論工作單

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1. **DO NOT extend the space provided.** Learn to select representative points and leave detailed explanations to your slides. You may adjust the layout within the designated page if needed.
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Part (a) What have you done

1. Major update for the drone system

- New power supply for the subsystem:** We changed the rover subsystem's power source from a power bank to direct power from the flight controller. Since each servo motor still consumes 700 to 800 mA of current during operation, we were concerned that this might affect the current distribution to the drone's main motors. Therefore, we decided to minimize the operation of the servo motors while the drone is in flight.
- Application of altitude hold mode:** We discovered that relying solely on throttle control could result in excessive descent speeds, potentially causing impact damage to the mechanism. Consequently, we decided to switch to altitude hold mode to control the drone, setting the maximum ascent and descent speeds to 50 cm/s.
- Wire organization:** We re-soldered and neatly organized all essential wiring. This improves the overall neatness of the cable routing and prevents the wires from getting entangled in the propellers during operation.
- Second drone for midterm test:** One week prior to the midterm test, we built an exact replica based on the first drone's design. This was to ensure we had a backup ready in the event of a crash during the midterm test.

2. Propeller thrust and stability analysis

- Experimental purpose and setup:** We conducted a motor thrust and propeller performance experiment to verify our propeller selection and understand the instability observed during previous flight tests. A thrust test bench was designed based on the drone arm structure, and the motor was controlled using the SpeedyBee flight controller with Betaflight and bidirectional D-shot to record throttle input, RPM, current, and thrust data.
- Testing process:** We tested six different propeller configurations, including 5040 four-blade, 5045 four-blade, 5040 three-blade, 5045 three-blade, racing three-blade, and 5045 two-blade propellers. Each propeller was tested from 5% to 60% throttle with 5% intervals, and the corresponding thrust, current, and RPM values were recorded.
- Main findings:** The results showed that the 5045 propellers generated higher thrust at the same RPM and reached the hovering thrust at a lower throttle level. However, they also had stronger vibration and a more sensitive thrust response, which may amplify instability during descent, especially under asymmetric ground effect.
- Final decision:** Based on the thrust-to-RPM, thrust-to-current, and efficiency comparison, we decided to continue using the 5040 four-blade propellers. Although the 5045 four-blade propellers showed slightly better efficiency near hovering thrust, the 5040 four-blade propellers provide a more stable and predictable flight response for our current system.

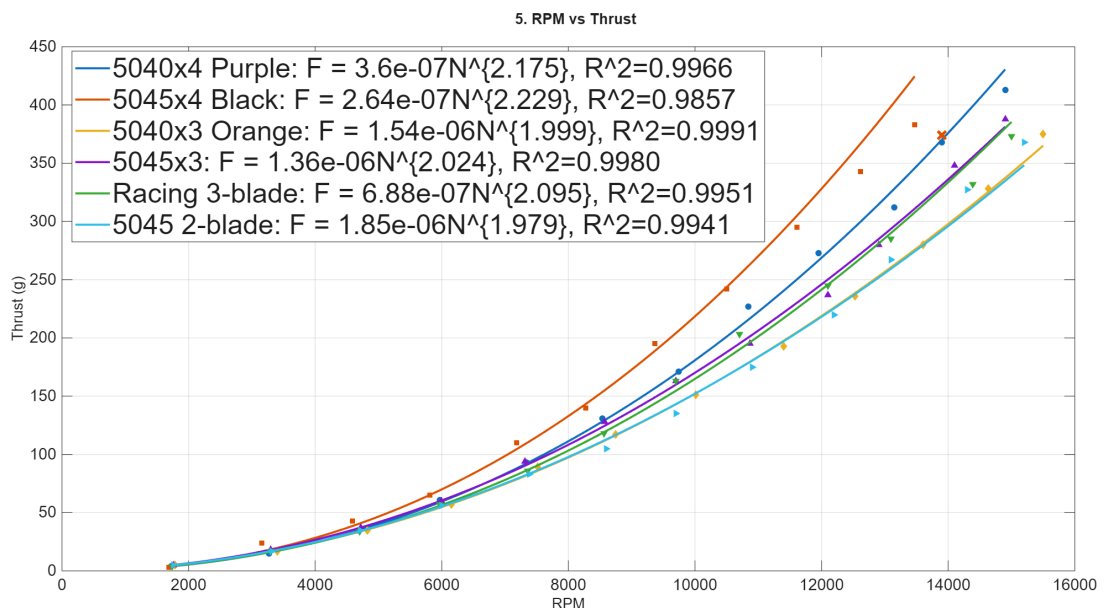


Figure 1: Thrust to RPM for six propellers

Part (b) What barriers you encounter:

1. **Balls against the wall:** Mechanical interference prevents the gate assembly from lowering effectively when attempting to capture targets positioned flush with vertical boundaries.

2. **Packet Flood caused by analog jitter**

The HW-504 joystick produced noisy analog readings. Even when the joystick was not intentionally moved, the `analogRead()` value still fluctuated slightly. In the original program, every small change in `leftCmd` or `rightCmd` was treated as a new command, so the ESP32 kept generating ESP-NOW packets almost every loop. This created a “packet flood,” especially when the joystick was continuously held in one direction. The solution was to add a joystick dead zone, command quantization, and a minimum sending interval. Instead of sending a packet for every 1-unit command fluctuation, the command is grouped into larger steps, so meaningless analog noise is ignored. This reduced unnecessary wireless traffic while keeping the vehicle controllable.

3. **ESP-NOW queue jam caused by blind fixed-interval sending**

The original transmitter sent packets at a fixed interval of 40 ms without checking whether the previous packet had actually finished sending. This could create a backlog inside the ESP-NOW stack when the joystick kept changing or when multiple operations were performed together. In practice, this made the vehicle feel delayed even though each individual packet was small. The solution was to avoid sending blindly. Instead, if a button is pressed or the joystick moves significantly, the ESP32 sends a packet immediately so that clamp and driving commands remain responsive. If no command changes, the transmitter still sends a heartbeat packet every 100 ms. This heartbeat is important because the ESP32-C3 receiver uses a 300 ms failsafe timeout: if no packet is received within that time, the receiver stops the wheels to prevent uncontrolled motion.

4. **Missing send callback control**

The initial ESP-NOW program did not use a send callback to confirm when the previous packet had completed. Without callback control, the ESP32 had no feedback about whether the wireless stack was ready for the next packet. This made the system vulnerable to packet congestion when the joystick, wheel commands, and clamp commands were operated at the same time. The improvement was to register an ESP-NOW send callback and use a `canSend` flag. After calling `esp_now_send()`, the program sets `canSend = false`. Only when the callback returns does it set `canSend = true`, allowing the next packet to be sent. This prevents old packets from piling up and makes the receiver respond to the most current operator input.

5. **Serial buffer blocking**

In the early version, both the ESP32 transmitter and ESP32-C3 receiver printed the clamp angle and wheel commands for almost every packet. When the loop generated packets faster than the serial port could transmit the text, the program had to wait for the serial buffer to clear. This blocking behavior slowed down the whole control loop. Serial logging is still useful for testing, but it should not be printed for every packet when evaluating actual response speed. Frequent serial printing is identified as one of the main causes of the 1–2 second delay.

Part (c) What is your next step:

1. Major upgrade of our ball gripping mechanism

Initial flight evaluations indicate a mission completion window of 50 to 60 seconds for the full transport cycle. Analysis of these results suggests that increasing drone velocity offers diminishing returns; reducing a 5-second transit to 4 seconds introduces disproportionate operational risk. Consequently, optimization efforts will focus on two primary avenues: minimizing the total number of required sorties and significantly decreasing the current 10-second individual retrieval latency.

- a. **2 or 3 balls at once:** Redesigning the gripper to carry multiple balls simultaneously will reduce sorties by 50-66%, significantly cutting mission time.
- b. **More powerful car motor:** Upgrading drive motors will provide the torque needed for rapid maneuvers and acceleration, minimizing retrieval latency.
- c. **Laser position indication:** A low-power laser will project an alignment guide, providing visual reference to reduce manual alignment time and improve target approach.
- d. **Faster ball release:** Replacing complex maneuvers with a gravity-assisted or actuation mechanism to ensure instantaneous, reliable drop of ball without landing the drone.

2. Finalizing our machine and construct the SOP for the final exam

While a comprehensive overhaul of the ball capture system remains a priority, our immediate focus involves locking the current mechanical configuration and establishing a rigorous SOP to guarantee repeatable performance during the final evaluation.

- a. **Final optimization:** The existing collection system lacks sufficient vertical elevation, causing the airframe to be displaced upward when approaching targets. While specialized pilot maneuvers currently mitigate this interference, we are developing a mechanical resolution to ensure the ball-capture assembly's vertical travel is fully decoupled from the main airframe.
- b. **SOP development:** Create a detailed Standard Operating Procedure for pre-flight checks, mission execution, and maintenance to ensure consistent performance in the final demonstration.

3. Approach the drone phenomena with a more analytical approach

Initial flight evaluations reveal significant positional instability and vertical oscillation during platform approach. To address this, we are shifting toward a rigorous analytical methodology to quantify the precise aerodynamic interference and fluid dynamic behaviors occurring near the landing surface.

4. Cost savings and structural optimization for mass production

Although we already achieved a robust solution for this task, we still aim to reduce the system's material and manufacturing complexity to facilitate a path toward high-volume production. This includes evaluating the possibility of replacing custom-machined components with off-the-shelf parts, optimizing the drone frame's geometry for faster additive manufacturing, and minimizing the total number of unique fasteners used across the assembly.



Figure 2: Thrust test bench for future experiments

List your reference in this page:

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